NEET-UG Botany | Chapter 12

ECOSYSTEM

NCERT Class 12 | Complete Chapter Analysis & Revision

Strictly Aligned with the NEET-UG Syllabus

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1️⃣ Chapter Overview: Ecosystem

NCERT Syllabus Mapping

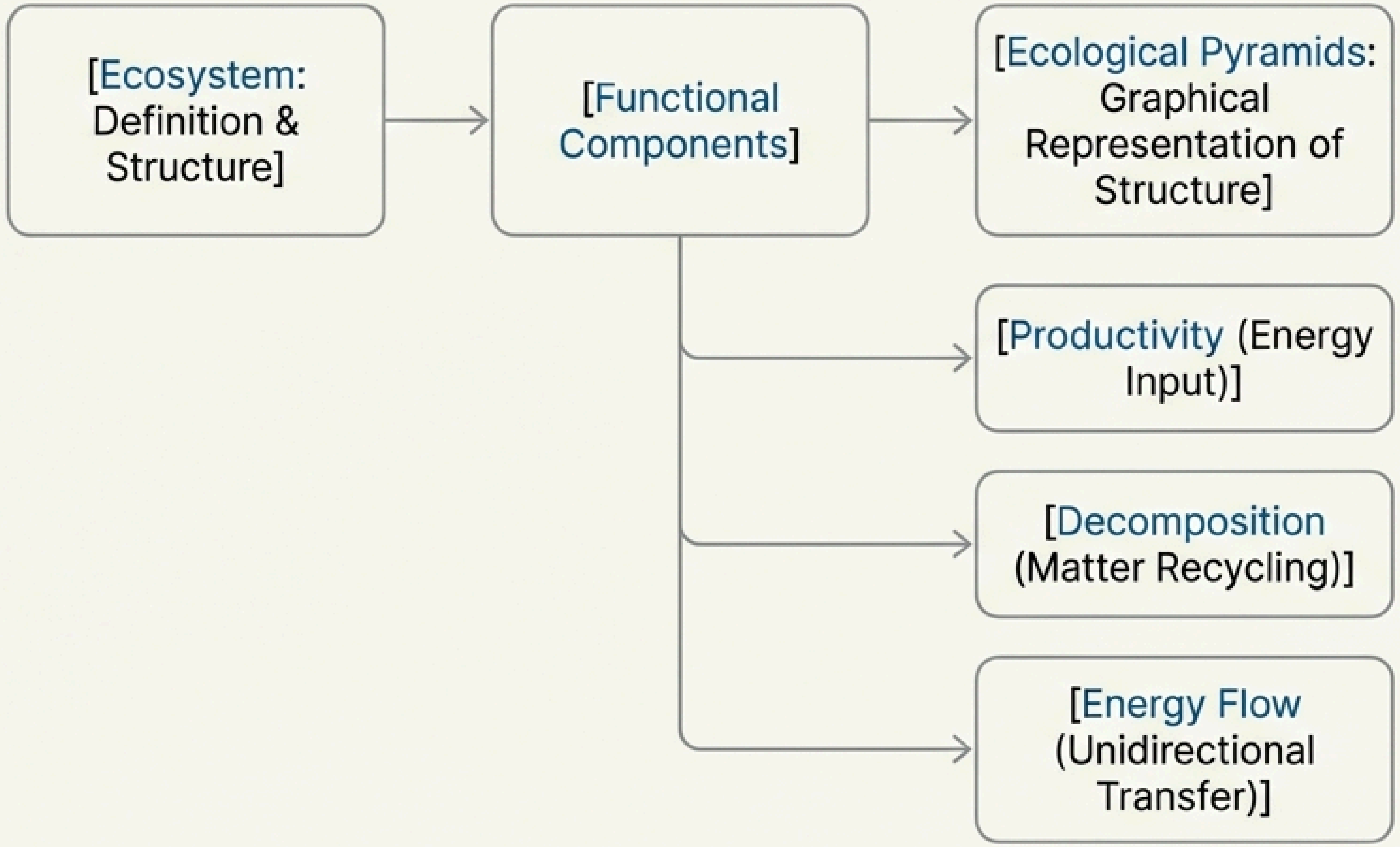
- **Class:** 12
- **Unit:** X (Ecology)
- **Chapter:** 12

NEET Weightage Trend

HIGH

(Ecology is a consistently high-yield unit in NEET-UG, and this chapter is fundamental.)

High-Level Concept Flow



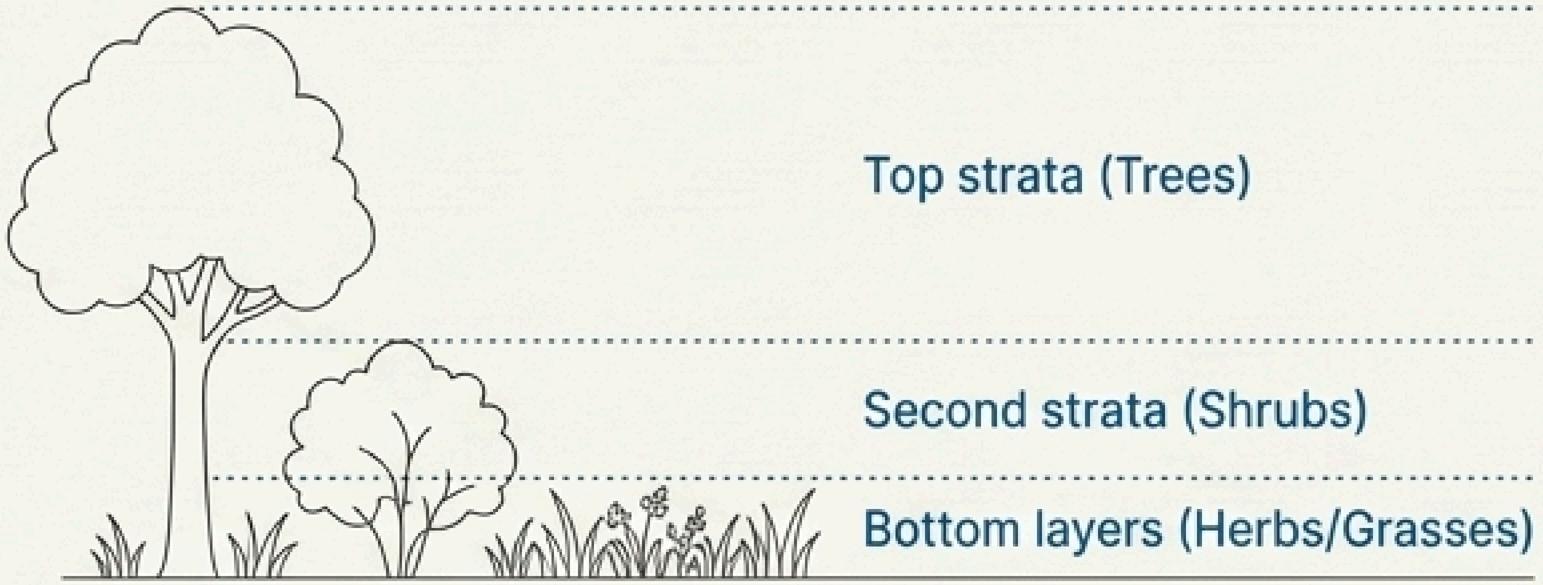
2️⃣ 12.1 Ecosystem – Structure and Function

Definition

- **Ecosystem:** A functional unit of nature, where living organisms interact among themselves and also with the surrounding physical environment.
- **Categories:** Can be divided into Terrestrial (e.g., forest, grassland, desert) and Aquatic (e.g., pond, lake, river). Man-made ecosystems include crop fields and aquariums.

Structural Features

- **Species Composition:** The identification and enumeration of plant and animal species.
- **Stratification:** The vertical distribution of different species occupying different levels.



The Four Functional Aspects

1. **Productivity**
2. **Decomposition**
3. **Energy Flow**
4. **Nutrient Cycling**

2️⃣ 12.2 Productivity

Key Definitions

- **Primary Production**: The amount of biomass or organic matter produced per unit area over a time period by plants during photosynthesis.
  - Units: weight (g^{-2}) or energy ($kcal\ m^{-2}$)
- **Productivity**: The *rate* of biomass production.
  - Units: rate ($g^{-2}\ yr^{-1}$) or ($kcal\ m^{-2}\ yr^{-1}$)

Types of Primary Productivity

$$NPP = GPP - R$$

- **Gross Primary Productivity (GPP)**: The rate of production of organic matter during photosynthesis.
- **Respiration losses (R)**: A considerable amount of GPP is utilised by plants in respiration.
- **Net Primary Productivity (NPP)**: The available biomass for consumption by heterotrophs (herbivores and decomposers).

Secondary Productivity

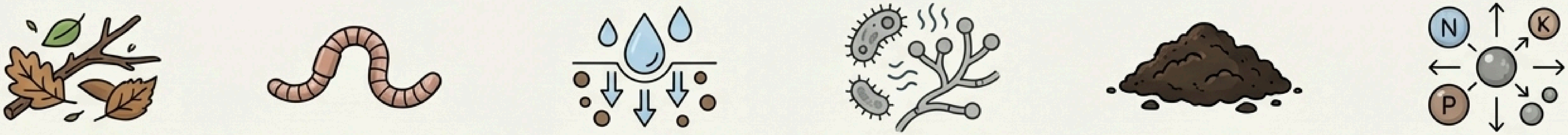
- The rate of formation of new organic matter by consumers.

2 12.3 Decomposition (Part 1: The Process)

Definition

- **Decomposition:** The process where decomposers break down complex organic matter into inorganic substances (CO_2, water, nutrients).
- **Detritus:** The raw material for decomposition (dead plant/animal remains, fecal matter).

The Five Key Steps in Decomposition



Detritus → **1. Fragmentation** → **2. Leaching** → **3. Catabolism** → **4. Humification** → **5. Mineralisation**

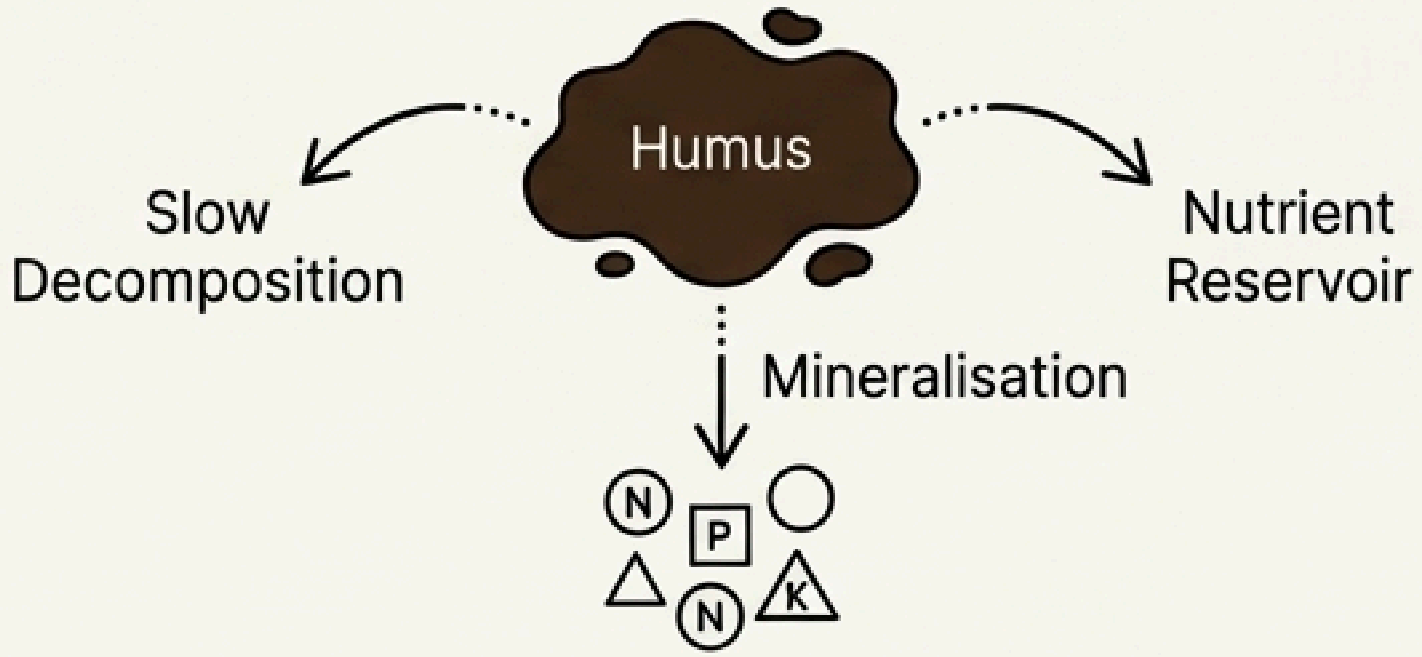
- **1. Fragmentation** (Breakdown by detritivores, e.g., earthworm)
- **2. Leaching** (Water-soluble nutrients move down into soil)
- **3. Catabolism** (Bacterial & fungal enzymes degrade detritus)
- **4. Humification** (Accumulation of humus)
- **5. Mineralisation** (Release of inorganic nutrients)

Note: Steps 1, 2, and 3 operate simultaneously on the detritus.

2️⃣ 12.3 Decomposition (Part 2: Products & Factors)

Products of Decomposition

- **Humification:** Leads to the accumulation of **humus**—a dark-coloured, amorphous substance.
  - **Properties**: Highly resistant to microbial action, decomposes slowly, colloidal nature, serves as a nutrient reservoir.
- **Mineralisation**: The process where humus is further degraded by microbes to release inorganic nutrients.



Factors Regulating Decomposition Rate

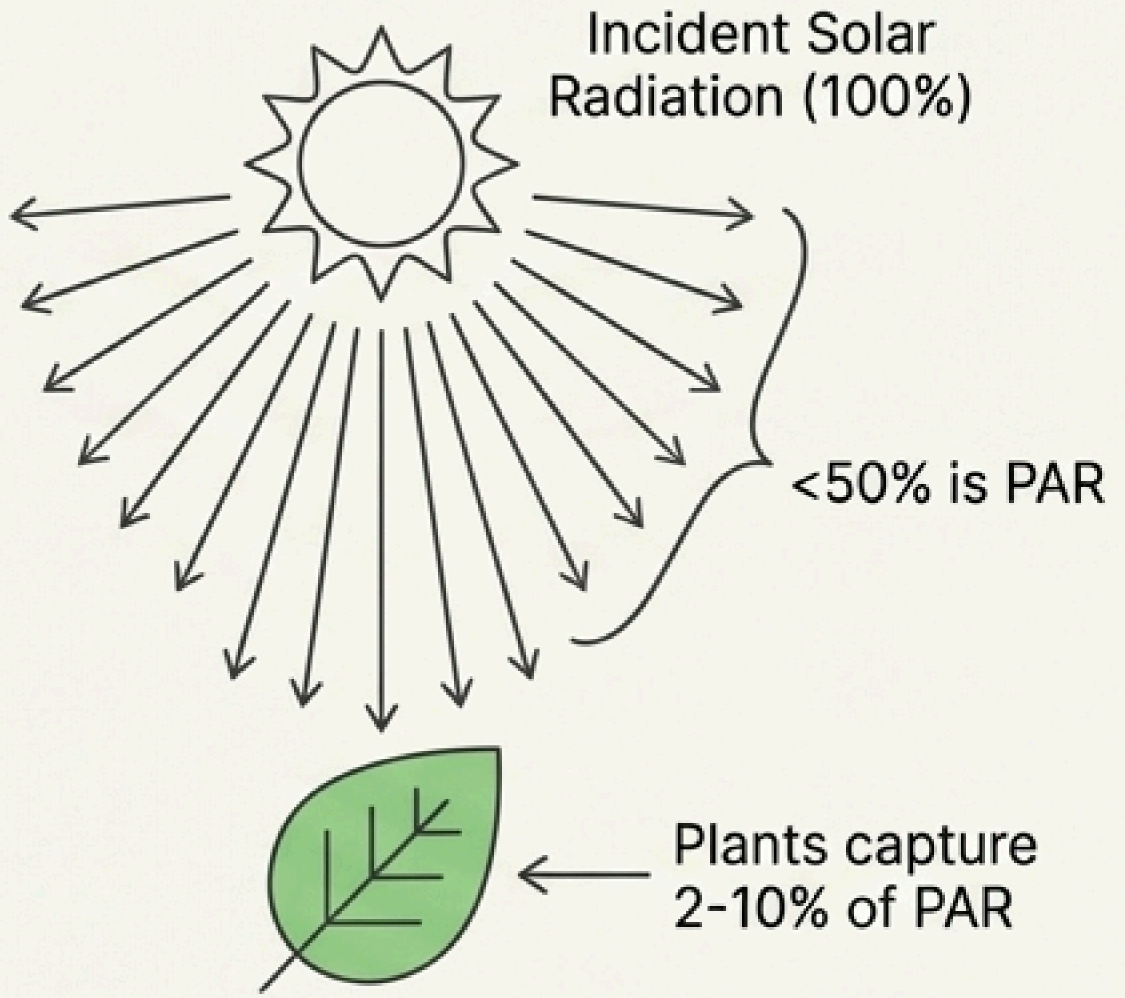
- **Chemical Composition of Detritus**:
  - 🔻 **Slower**: If detritus is rich in lignin and chitin.
  - ▲ **Quicker**: If detritus is rich in nitrogen and water-soluble substances like sugars.
- **Climatic Factors (Most Important)**:
  - ▲ **Favours**: Warm & moist environment.
  - 🔻 **Inhibits**: Low temperature & anaerobiosis (leads to build-up of organic materials).

***Note**: Decomposition is largely an oxygen-requiring process.*

2️⃣ 12.4 Energy Flow (Part 1: Fundamentals & Trophic Levels)

Energy Source & Capture

- **Primary Source**: Sun (except for deep-sea hydrothermal ecosystems).
- **Photosynthetically Active Radiation (PAR)**: Less than 50% of incident solar radiation.
- **Capture Efficiency**: Plants capture only 2-10% of PAR to sustain the entire living world.



Core Principles

- **Unidirectional Flow**: Energy flows from the sun to producers and then to consumers.
- **Thermodynamics**: Ecosystems are not exempt from the First and Second Laws. They require a constant energy supply to counteract entropy.

Trophic Levels

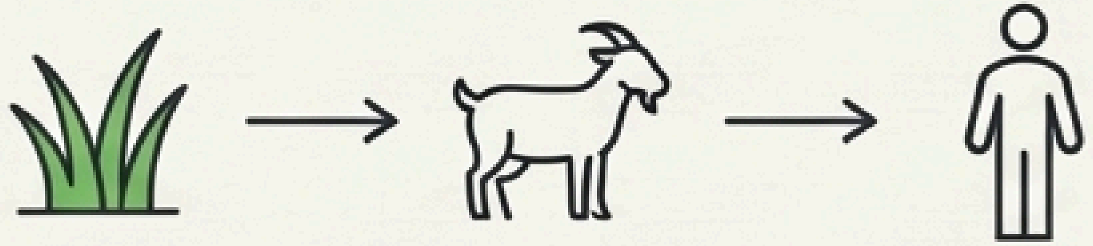
The specific place an organism occupies in a food chain based on its food source.

- **T1**: Producers (Plants)
- **T2**: Primary Consumers (Herbivores)
- **T3**: Secondary Consumers (Primary Carnivores)
- **T4+**: Tertiary Consumers, etc.

2️⃣ 12.4 Energy Flow (Part 2: Food Chains & Webs)

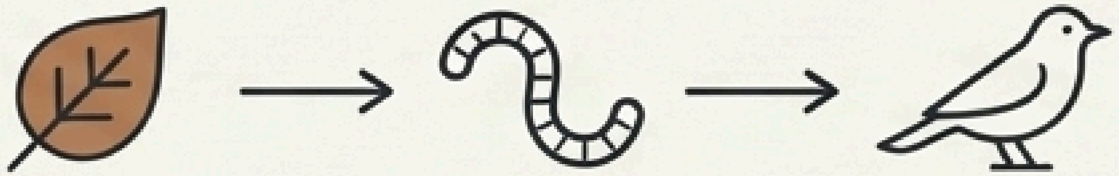
Grazing Food Chain (GFC)

- Begins with producers (plants).
- Primary consumers are herbivores.
- *This is the major conduit for energy flow in **aquatic ecosystems**.*



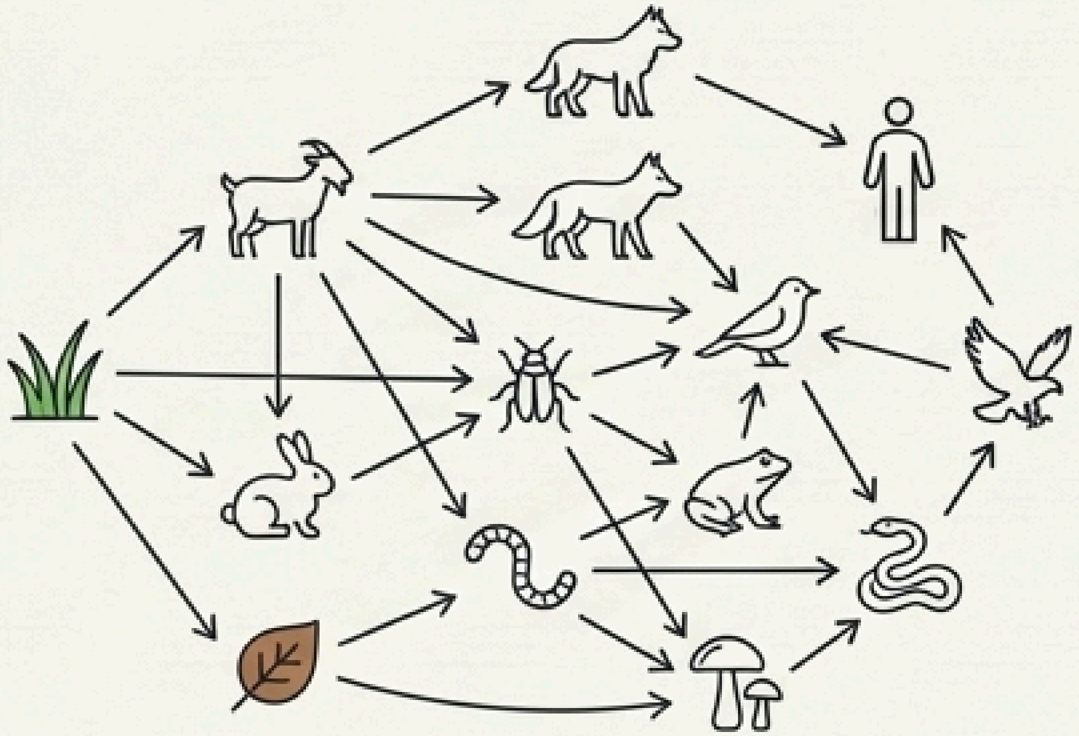
Detritus Food Chain (DFC)

- Begins with dead organic matter (detritus).
- Involves decomposers (saprotrophs) like fungi and bacteria.
- *A much larger fraction of energy flows through the DFC in **terrestrial ecosystems**.*



Key Concepts

- **Food Web**: Formed by the natural interconnection of food chains.
- **Standing Crop**: The mass of living material at a particular trophic level at a particular time.
  - Measured as biomass (mass of living organisms) or number per unit area.
  - Measurement in terms of **dry weight is more accurate.**



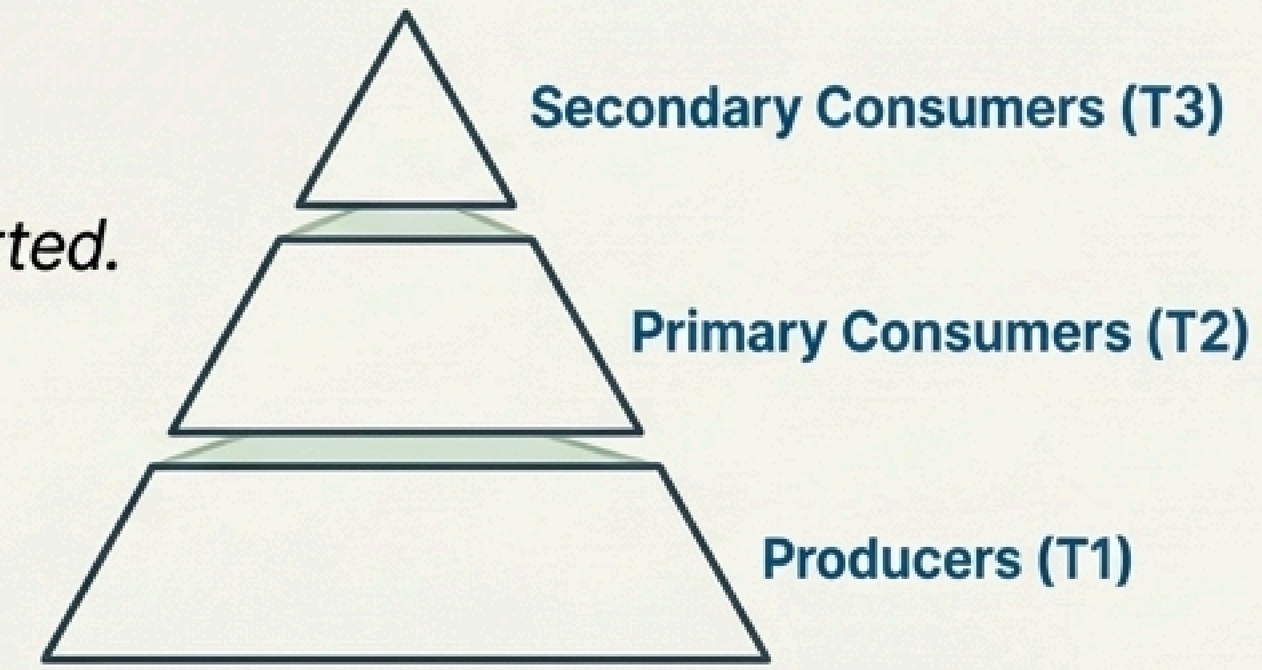
2️⃣ 12.5 Ecological Pyramids

Definition

- A graphical representation of the relationship between organisms at different trophic levels in terms of number, biomass, or energy.
- **Structure:** The base always represents the producers (first trophic level), and the apex represents the tertiary or top-level consumer.

Three Types of Ecological Pyramids

1. **Pyramid of Number:** Represents the number of individual organisms at each trophic level. *Usually upright, but can be inverted.*
2. **Pyramid of Biomass:** Represents the total mass of living organisms (biomass) at each trophic level. *Usually upright, but can be inverted.*
3. **Pyramid of Energy:** Represents the energy content at each trophic level. ****Always upright.****

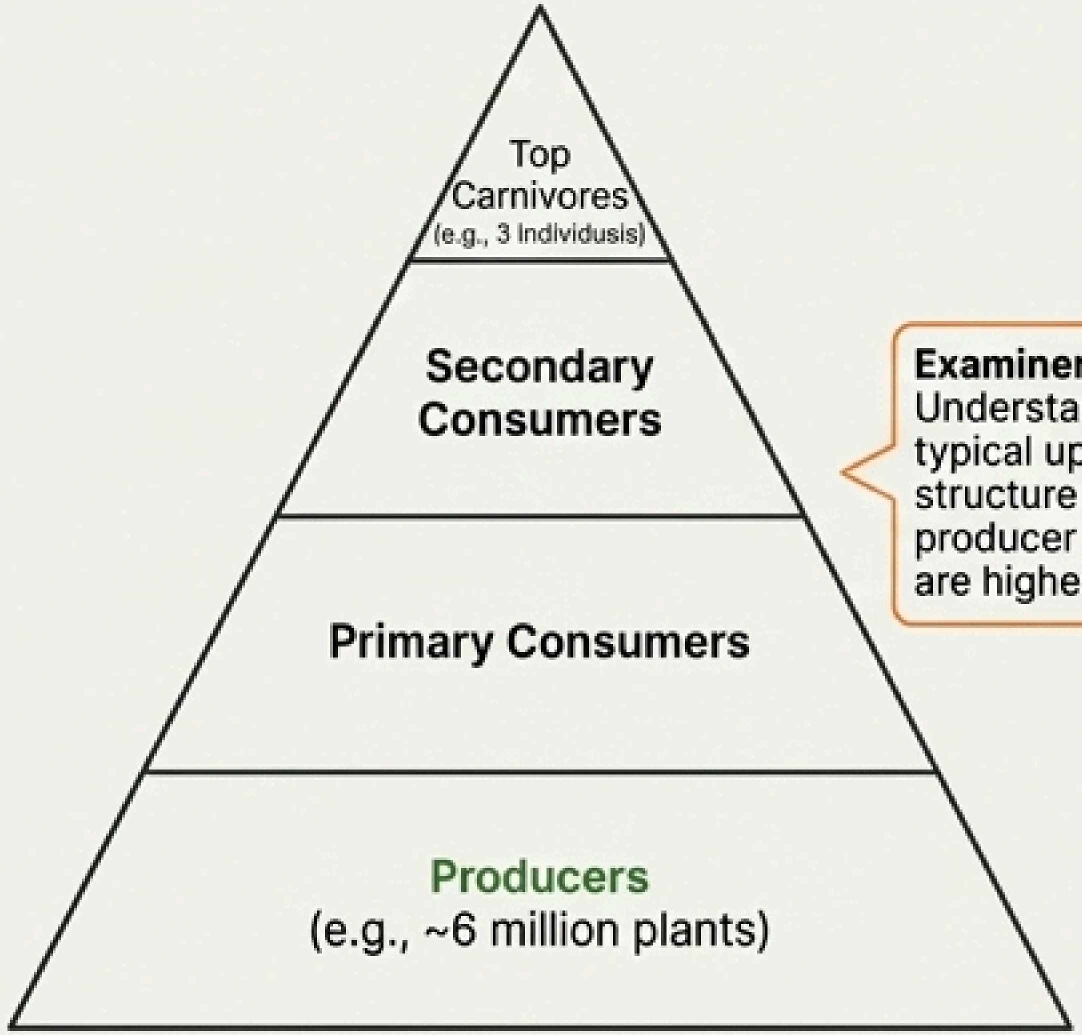


Limitations

- Assumes a simple food chain (not a food web).
- Does not accommodate species that occupy multiple trophic levels.
- Saprophytes (decomposers) are not given any place.

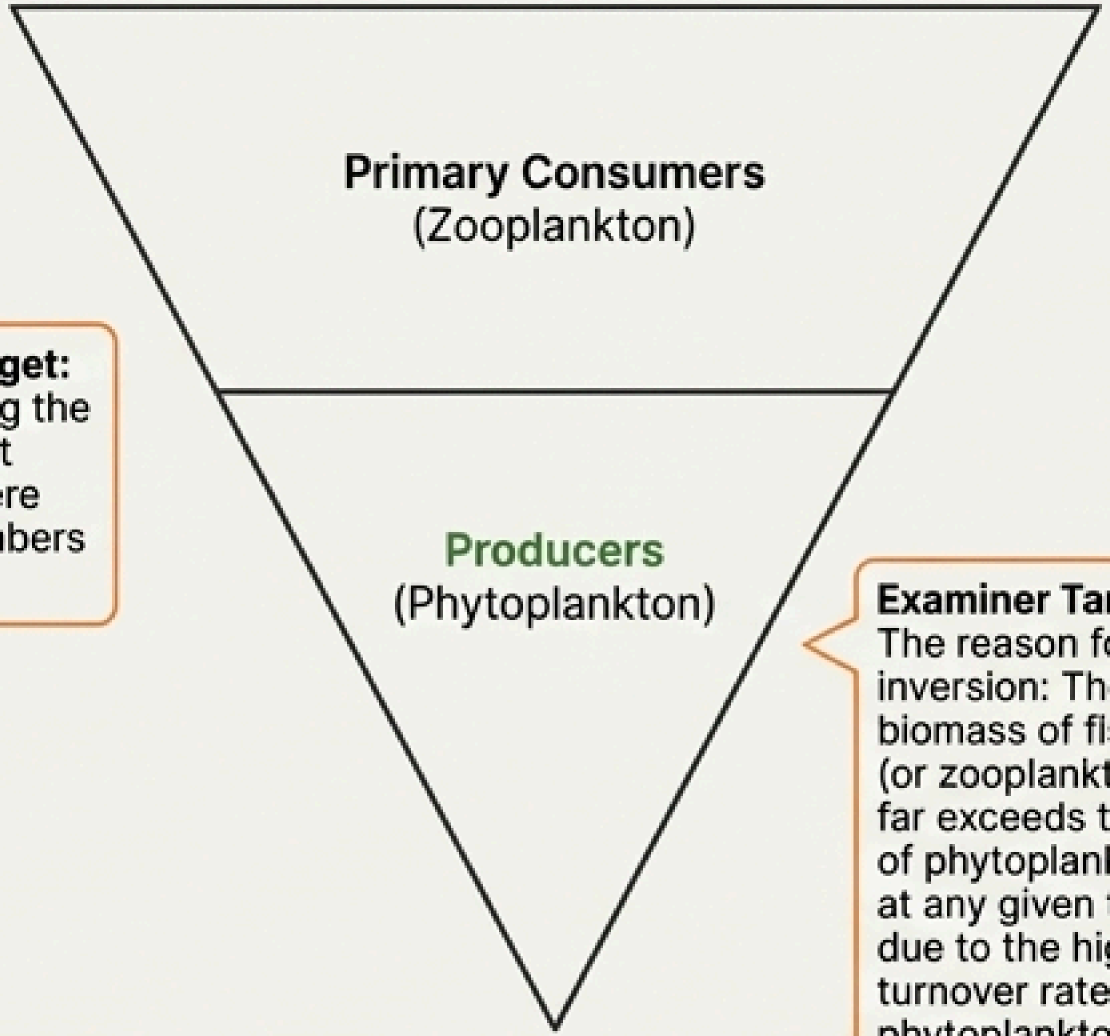
4 4️⃣ Diagrams Focus: Ecological Pyramids

A: Pyramid of Number (Grassland Ecosystem)



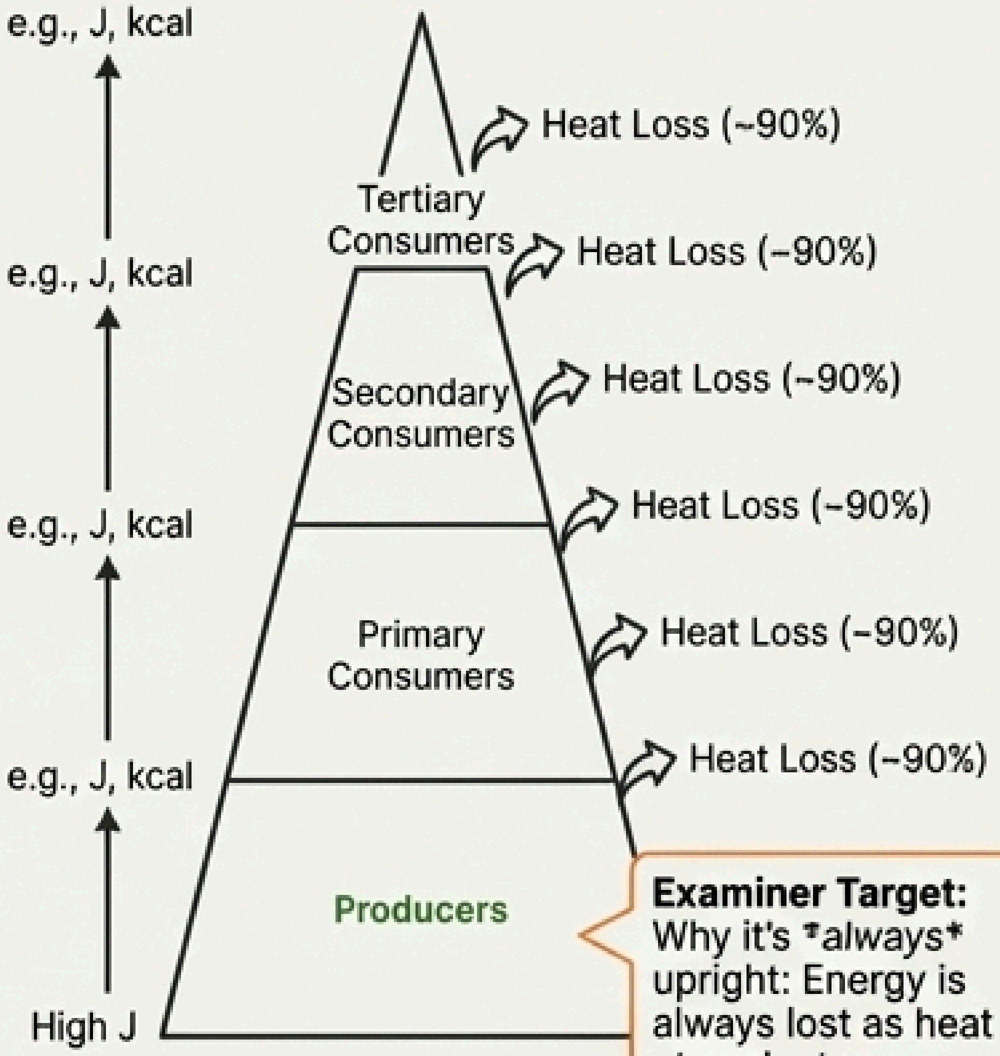
Examiner Target: Understanding the typical upright structure where producer numbers are highest.

B. Inverted Pyramid of Biomass (Aquatic Ecosystem)



Examiner Target: The reason for inversion: The biomass of fishes (or zooplankton) far exceeds that of phytoplankton at any given time due to the high turnover rate of phytoplankton.

C. Pyramid of Energy



Examiner Target: Why it's *always* upright: Energy is always lost as heat at each step according to the second law of thermodynamics. It can never be inverted.

3 NCERT Line Emphasis

⚠ NCERT Line

"Net primary productivity is the available biomass for the consumption to heterotrophs (herbiviores and decomposers)."

★ Direct NEET Pickup

"The annual net primary productivity of the whole biosphere is approximately **170 billion tons** (dry weight) of organic matter. Of this, despite occupying about 70 per cent of the surface, the productivity of the oceans are only **55 billion tons**."

⚠ NCERT Line

"In a terrestrial ecosystem, a much larger fraction of energy flows through the **detritus food chain** than through the GFC."

★ Direct NEET Pickup

"The transfer of energy follows **10 per cent law** – only 10 per cent of the energy is transferred to each trophic level from the lower trophic level."

5️⃣ Comparison: Grazing Food Chain (GFC) vs. Detritus Food Chain (DFC)

Parameter	Grazing Food Chain (GFC)	Detritus Food Chain (DFC)
Starting Point	**Producers** (Living green plants)	Dead organic matter (**Detritus**)
Energy Source	Solar Radiation	**Chemical energy** stored in detritus
Dominant Ecosystem	**Aquatic Ecosystems**	**Terrestrial Ecosystems**
Magnitude of Flow	Major conduit for energy flow	A much larger fraction of energy flows through it
Organisms	Producers, Herbivores, Carnivores	**Detritivores**, **Saprotrophs** (Fungi, Bacteria)

6 NEET-UG Focus: Common Traps & Key Concepts

Previous Year Question Themes

- Calculation of energy transfer using the **10% law.**
- Identifying the shape of pyramids for specific ecosystems (e.g., pyramid of biomass in sea, pyramid of numbers on a tree).
- Factors affecting the rate of decomposition (Lignin, Chitin, Temperature, Moisture).
- The definition of **PAR** and the percentage captured by plants (2-10%).
- The global NPP data (170 vs. 55 billion tons).

Common Traps & Misconceptions

- **Trap:** Assuming the Pyramid of Biomass is always upright.
  - **Clarification:** It is **inverted** in the sea.
- **Trap:** Confusing 'Standing Crop' with 'Productivity'.
  - **Clarification:** Standing Crop is the biomass *at a given time* (a snapshot). Productivity is the *rate* of production *over time*.
- **Trap:** A species belongs to only one trophic level.
  - **Clarification:** A species can occupy multiple trophic levels (e.g., a sparrow eating seeds vs. insects).

7️⃣ Assertion–Reason & Statement Analysis

Assertion-Reason Section

1. **Assertion (A):** The pyramid of biomass in sea is generally inverted.
   Reason (R): The biomass of fishes far exceeds that of phytoplankton.
   - **Analysis:** Both A and R are true, and R is the correct explanation of A.
2. **Assertion (A):** Decomposition is slower if detritus is rich in lignin and chitin.
   Reason (R): Warm and moist environments inhibit decomposition.
   - **Analysis:** A is true, but R is false (warm and moist ***favour*** decomposition).

Conceptual Checks (True/False)

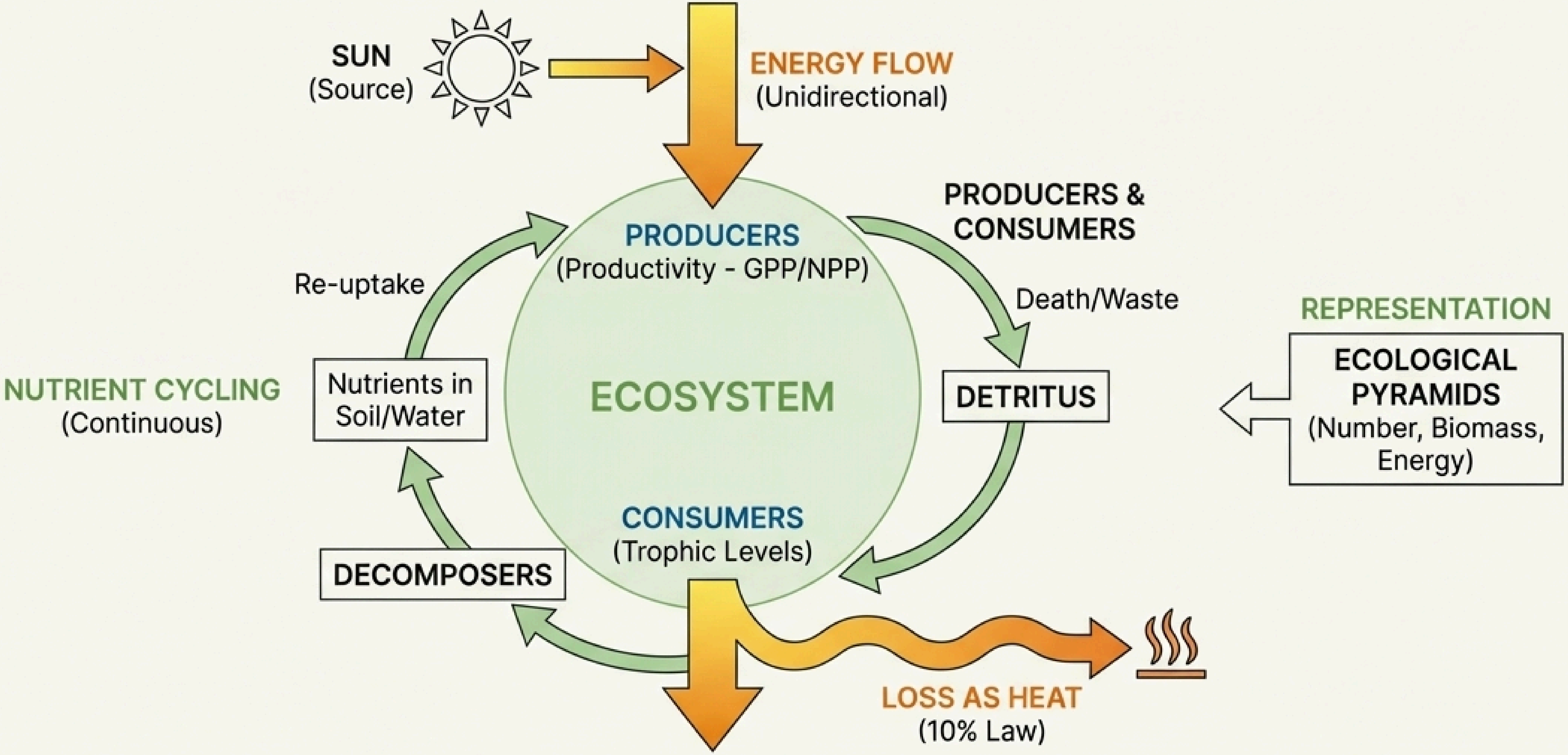
- **Statement 1:** Saprophytes are given a specific place at the base of ecological pyramids.
  - **False.** This is a major limitation; they are not given any place in the pyramids.
- **Statement 2:** In an aquatic ecosystem, the grazing food chain (GFC) is the major conduit for energy flow.
  - **True.** This is a key distinction from terrestrial ecosystems.

8 Rapid Revision: Ecosystem at a Glance

- **Structure**: Species Composition & Stratification (vertical layering).
- **Function**: Productivity, Decomposition, Energy Flow, Nutrient Cycling.
- **Productivity**: Rate of biomass production. Key formula: **NPP = GPP - R**.
- **Decomposition**: Breakdown of detritus. Key Factors: Oxygen, Temperature, Moisture, Chemical Composition (Lignin/Chitin slow it down).
- **Energy Flow**: Unidirectional. Follows the **10% law**. A large fraction is lost as heat.
- **Food Chains**: **GFC** is major in aquatic systems. **DFC** is major in terrestrial systems.
- **Ecological Pyramids**:
  - **Energy**: Always Upright.
  - **Biomass**: Inverted in the Sea.
  - **Number**: Inverted on a single Tree ecosystem.
- **Key Data**: PAR is <50% of solar radiation. Plants capture 2-10% of PAR. Global NPP is ~170 billion tons; Oceans contribute only 55 billion tons.

Keywords List: Stratification, PAR, GPP, NPP, Detritus, Humification, Mineralisation, Catabolism, Standing Crop, 10%Law, Food Web, Saprotrophs.

9 Final Chapter Snapshot: The Core Logic



If you remember only this much, you can solve most questions:
Energy **flows** one way and is lost as heat. Nutrients **cycle** continuously, driven by decomposition.
The Pyramid of Energy reflects this one-way flow and is therefore **always upright**.